

SHARMILA DUPPALA

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EDUCATION

University of Maryland, College Park

Aug. 2019 - Dec 2025

Ph.D. in Computer Science

Advisor(s): [John P. Dickerson](#), [Aravind Srinivasan](#)

Dissertation: Probabilistic Notions of Fairness in Machine Learning and Combinatorial Optimization

Stony Brook University

Aug. 2017 - Dec. 2019

Masters in Computer Science

Advisor: [Rezaul Chowdhury](#)

Dissertation: Optimizing Parallel Systems Employing Reducers

National Institute of Technology, Surat

Aug. 2012 - Dec. 2016

Bachelors in Computer Science

KEY PROJECTS AND EXPERIENCE

Interpretability of Language Reasoning Tasks in LLMs, DoD

Dec. 2024 – Present

Research Assistant([Github Repo](#))

[Dr. Erik Krokos](#), [Prof. Aravind Srinivasan](#)

- Developed a framework that maps the subtasks within a syntactic reasoning task to specific attention heads in Transformer models by leveraging sparse attention to identify heads most likely responsible for each subtask.
- Performed causal mediation analysis (via activation & path patching) to trace information pathways among candidate heads that constitute the reasoning circuit for the downstream task.
- Implemented and empirically evaluated post-training algorithms (PPO, GRPO, DPO) on state-of-the-art pre-trained and instruction fine-tuned models to compare their effects on interpretability.

Evaluating VLMs on Event Sequence Visualization images

Fall. 2024

Published in [EMNLP 2025](#)

- Developed a benchmark to evaluate how visualization properties such as structure type (linear, tree, and graph) and structural density affect Vision Language Model's (VLM) performance on event sequence data.
- Curated a dataset with expert-validated Q/A pairs on process visualizations mined from sequence data for two categories of tasks: *data extraction* and *question answering*; and evaluated on the frontier VLMs and reported limitations such as, high edge density increased hallucination rates and model's performance dropped with increased structural complexity across most models.

Distantly Supervised Customer Transcript Labeling, Optum Labs June. 2023 – Aug. 2023

Applied Scientist Intern

[Dr. Carlos W. Morato](#)

- Designed and developed a semi-supervised learning model to segment and classify multi-labeled conversational data.
- Representation learning was used to model semantic topics as distinct clusters followed by a custom ranking algorithm to accurately map unstructured dialogue segments to the most relevant topics.

SELECTED PUBLICATIONS

Visit my [Google Scholar](#) for complete list

- **ProcVQA: Benchmarking VLM Performance on Mined Process Visualizations**

[EMNLP](#), 2025

K. Zinat, S. Abrar, S. Saha, **S. Duppala**, S. Sakhamuri, Z. Liu

- **Robust Fair Clustering with Noisy Memberships**
S. Duppala, J. Luque, J. Dickerson, S. Esmaili

[AISTATS](#), 2025

- **Randomized Rounding for Proportional Fair Matching** AAAI (*Oral*), 2025
S. Duppala, N. Grammel, J. Luque, C. MacRury, A. Srinivasan
- **Concentration of Submodular Functions Under Negative Dependence** ITCS, 2025
^{abc}S. Duppala, G. Z. Li, J. Luque, R. Valieva, A. Srinivasan
- **Barter Exchange with Shared Item Valuations** TheWebConference, 2024
J. Luque, S. Duppala, J. Dickerson, A. Srinivasan
- **Group Fairness in Set Packing Problems** IJCAI 2023
S. Duppala, J. Luque, J. Dickerson, A. Srinivasan
- **Rawlsian Fairness in Online Bipartite Matching** AAAI, 2023
S. Esmaeili, S. Duppala, V. Nanda, J. Dickerson, A. Srinivasan
- **Fair labelled Clustering** KDD, 2022
S. Esmaeili, S. Duppala, B. Brubach, J. Dickerson
- **Improved MapReduce Load Balancing through Distribution-Dependent Hash Function Optimization** ICPADS, 2020
S. Duppala*, Z. Ahmad*, R. Chowdhury, S. Skiena
- **Data races and the discrete resource-time tradeoff problem with resource reuse over paths** SPAA, 2019
R. Das, S. Tsai, S. Duppala, J. Lynch, E. Arkin, R. Chowdhury, J. Mitchell, S. Skiena
- **Simultaneous Individual and Group Fairness in Set Cover** *Under Submission*, 2025
S. Duppala, N. Grammel, A. Srinivasan
- **Asymmetric Barter Exchange** *Under Submission*, 2025
J. Luque, S. Duppala, J. Dickerson, A. Srinivasan
- **Interpretability of Language Reasoning Tasks in LLMs via Sparse Attention.** *Working paper*
S. Duppala, E. Krokos, A. Srinivasan

TECHNICAL SKILLS

- **Libraries/Tools for Machine Learning:** Numpy, Pandas, Scikit-learn, PyTorch, Transformer-Lens, HuggingFace Transformers, Datasets, and Tokenizers, TRL, Kubernetes.
- **Mathematical Optimization:** Gurobi, NetworkX
- **Programming Languages:** C, C++, Python, Java

KEY COURSEWORK

- **Machine Learning:** Discrete Probability and Statistics, Multimodal Foundational Models, Advanced Numerical Optimization, Linear Algebra, Deep Learning Theory.
- **Algorithms:** Mechanism Design, Randomized Algorithms, Computational Geometry.

POSITIONS OF RESPONSIBILITY

- **Program Committee Member; Reviewer; Sub-reviewer** May. 2018 – Present
AAAI 2026, SPAA, ESA 2025, WebConf. 2024, 2025, 2026, ICLR 2023, SODA 2019, 2023
- **Capital Area Theory Seminar & Alg. Fairness Seminar** Aug. 2019 – May. 2025
Responsible for co-organizing CS theory weekly seminar hosting external speakers and a reading group discussing the latest papers in fairness literature.
- **Graduate Teaching Assistant** Aug. 2019 – May. 2025
Natural Language Processing, Advanced Algorithms, Computer Systems, and Discrete Structures.
- **Curriculum Designer and Instructor, Girls Talk Math** May. 2021 – Aug. 2021
Responsible for designing curriculum on undergraduate mathematics topics like Group Theory, Network Theory and Quantitative Finance for high school students.